DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 2 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 2

Subject Code & Name: 24AF1000BS201 - ENGINEERING MATHEMATICS – II

Time : 3 Hours] **[Total Marks : 60**

Instructions to the Students:

1.Each question carries 12 marks.
2.Question No. 1 will be compulsory and include objective-type questions.
3.Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4.The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5.Use of non-programmable scientific calculators is allowed.
6.Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question) 12

1 If $z = -1 - i$, then its amplitude is

(a) $\frac{\pi}{4}$ (b) $-\frac{\pi}{4}$

(c) $\frac{5\pi}{4}$ (d) $-\frac{5\pi}{4}$.

2 If $z = \cos\theta + i\sin\theta$, then the value of $z^n - \frac{1}{z^n}$ is

(a) $2i\sin n\theta$ (b) $2i\cos n\theta$

(c) $\sin n\theta$ (d) $\cos n\theta$.

3 The module of $(\sqrt{i})^{\sqrt{i}}$ is equal to

(a) $e^{\frac{\pi}{4\sqrt{2}}}$ (b) $e^{\frac{\pi}{2\sqrt{2}}}$

(c) $e^{-\frac{\pi}{4\sqrt{2}}}$ (d) $e^{-\frac{\pi}{2\sqrt{2}}}$.

4 The integrating factor of the differential equation $\frac{dy}{dx} + y\cos x = \frac{\sin 2x}{2}$ is

(a) $e^{\sin x}$ (b) $e^{\sin^2 x}$

(c) $e^{\sin 2x}$ (d) $e^{\cos x}$

5 The general solution of $\frac{1}{x^2y^2}(y\,dx + x\,dy)$ is

(a) $x + y = c$

(b) $x - y = c$

(c) $xy = c$

(d) $\frac{1}{x^2y^2}$

6 The solution of $(D^2 - 2D + 1)y = 0$ is

(a) $(c_1 + c_2x)e^x$

(b) $(c_1 + c_2x)e^{-x}$

(c) $(c_1 - c_2x)e^x$

(d) $c_1e^x + c_2\,e^{-x}$

7 The complimentary function of the differential equation $y'' - 3y' + 2y = 12$ is

(a) $c_1e^x + c_2e^{2x}$

(b) $c_1e^x + c_2e^{-2x}$

(c) $c_1e^{-x} + c_2e^{2x}$

(d) $c_1e^{-x} + c_2e^{-2x}$

8 The particular integral of the differential equation $(D^2 - 4)y = \sin 3x$ is

(a) $-\frac{1}{13}\sin(3x)$

(b) $\frac{1}{13}\sin(3x)$

(c) $\frac{1}{5}\sin(3x)$

(d) $-\frac{1}{5}\sin(3x)$

9 If $f(x) = x^2$ in $-2 < x < 2$, then the Fourier constant a_n is equal to

(a) $\int_0^2 x^2 \frac{\cos(n\pi x)}{2}\,dx$

(b) $\int_0^2 x^2 \frac{\sin(n\pi x)}{2}\,dx$

(c) $\int_0^2 x^2 \cos(nx)\,dx$

(d) $\int_0^2 x^2 \sin(nx)\,dx$

10 For the half-range sine series for $f(x) = 1$ in $(0, \pi)$ the value of b_n is

(a) $\frac{2(1+cos n\pi)}{n\pi}$

(b) $\frac{2(1-cos n\pi)}{n\pi}$

(c) $\frac{2(1+sin n\pi)}{n\pi}$

(d) $\frac{2(1-sin n\pi)}{n\pi}$

11 Div($\vec{r}$), where $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$, is equal to

(a) 0

(b) 1

(c) 3

(d) -3

12 The value of $\iint_S (yz\,dy\,dz + zx\,dz\,dx + xy\,dx\,dy)$, where S is the surface of the sphere $x^2 + y^2 + z^2 = 1$, is

(a) 0

(b) 4π

(c) $\frac{4\pi}{3}$

(d) π

Q2. Solve the following.

A) If $\alpha, \alpha^2, \alpha^3, \alpha^4$ are roots of the equation $x^5 - 1 = 0$, then show that: 6

$$(1-\alpha)(1-\alpha^2)(1-\alpha^3)(1-\alpha^4) = 5.$$

B) Prove that: $\tan\left[i \log \frac{a-ib}{a+ib}\right] = \frac{2ab}{a^2-b^2}$. 6

Q3. Solve the following.

A) Solve: $y\,dx - x\,dy + \log(x)\,dx = 0$. 6

B) Solve: $(1+y^2)dx + \left(x - e^{\tan^{-1} y}\right)dy = 0$. 6

Q4. Solve Any Two of the following.

A) Solve: $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 2y = e^x + \cos x$. 6

B) Solve the following differential equation by the method of variation of parameters: 6

$$y'' - 2y' + 2y = e^x \tan x.$$

C) Solve: $x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 5y = x^2 \sin(\log x)$. 6

Q5. Solve Any Two of the following.

A) Find the Fourier series expansion of $f(x) = 4 - x^2$ in the range $(0, 2)$. 6

B) Find the Fourier series for the function $f(x) = x^3$ in the interval $(-\pi, \pi)$. 6

C) Obtain the half-range Fourier cosine series for the function $f(x) = \pi - x$ in $0 \leq x \leq \pi$. 6

Q6. Solve Any Two of the following.

A) Find $\nabla \cdot \vec{F}$ and $\nabla \times \vec{F}$, where $\vec{F} = \text{grad}\,(x^3 + y^3 + z^3 - 3xyz)$. 6

B) If $\vec{r}$ is the position vector and r is its magnitude, then prove that: 6

(i) $\nabla \cdot (r^n \vec{r}) = (n+3)r^n$

(ii) $\nabla \times (r^n \vec{r}) = 0$.

C) Apply Stokes' theorem to evaluate $\int_C [4y\,dx + 2z\,dy + 6y\,dz]$, where C is the curve of intersection of $x^2 + y^2 + z^2 = 6z$ and $z = x + 3$. 6